

Lab: Motor Control and Active Inference

Learning Goal: Analyze motor systems through the Active Inference framework, mapping neural circuits to predictions, errors, and policy selection.

Part 1: Predictions Not Commands

Exercise: Compare the classical and Active Inference accounts of reaching for a coffee cup:

Step	Classical Motor Control Account	Active Inference Account
1	Motor cortex computes desired trajectory	Motor cortex generates proprioceptive prediction ("arm will be at cup location")
2	Motor cortex sends commands to spinal cord	Prediction descends to spinal cord as expected proprioceptive state
3	Spinal cord activates muscles	Spinal reflex arcs detect prediction error between predicted and actual arm position
4	Muscles contract in correct sequence	Reflex arcs contract muscles to fulfill proprioceptive prediction
5	Sensory feedback confirms movement	Prediction error decreases as arm approaches predicted position

Now apply the Active Inference account to: (a) catching a thrown ball, (b) speaking a sentence, (c) playing a piano scale. For each, identify the prediction, the prediction errors, and the reflex mechanisms that fulfill the prediction.

Write your response here

Part 2: Cerebellar Forward Model Analysis

Learning Goal: Understand how the cerebellum predicts motor consequences and updates through error.

Exercise: Consider a novice learning to throw darts:

1. **Early learning phase:** The cerebellar forward model is inaccurate. For each throw, describe:
 2. What the cerebellum predicts (expected landing position)
 3. What actually happens (actual landing position)
 4. The motor prediction error (climbing fiber signal)
5. How the forward model updates (parallel fiber-Purkinje LTD)
6. **Late learning phase:** The cerebellar forward model is accurate. How does the pattern change?
7. **What happens if the dart's weight suddenly changes?** Describe the re-learning process in terms of prediction errors and model updating.
8. **Why are cerebellar patients bad at timing but not at force?** Connect this to the cerebellum's role in temporal prediction.

Write your response here

Part 3: Basal Ganglia Policy Selection Simulation

Learning Goal: Map the direct/indirect/hyperdirect pathways to Active Inference policy selection.

Exercise: Consider choosing between two motor policies: (A) reach left for water, (B) reach right for coffee.

Scenario 1: You're thirsty (strong preference for water)

Pathway	Activity	Computational Effect
Direct (D1) to Policy A	HIGH	"Go" for water-reaching policy — selected policy gets high precision
Indirect (D2) to Policy B	HIGH	"No-go" for coffee-reaching — competing policy suppressed
Hyperdirect	LOW	No need to stop — no surprising events to trigger re-evaluation

Scenario 2: You hear a loud crash while reaching

Pathway	Activity	Computational Effect
Direct	SUPPRESSED	All ongoing policies halted
Indirect	SUPPRESSED	All ongoing policies halted
Hyperdirect	HIGH	Global stop signal — re-evaluate all policies because surprising event

Now design Scenario 3: You are equally thirsty and caffeinated. Describe the pathways and the ACC's role in resolving conflict.

Write your response here

Part 4: Sensory Attenuation Experiments

Learning Goal: Understand self-generated sensation attenuation as evidence for forward models.

Exercise 1 — The Tickling Thought Experiment:

You touch your own arm with your right hand. Then a friend touches the same spot with the same pressure.

Condition	Forward Model Prediction	Actual Sensation	Prediction Error	Perceived Intensity
Self-touch	"Touch will occur at X, pressure Y"	Touch at X, pressure Y	Very small	Low (attenuated)
Other-touch	No prediction (unpredicted)	Touch at X, pressure Y	Large	High (full intensity)

Explain why the self-generated touch is perceived as less intense. What does this reveal about motor predictions?

Exercise 2 — Force Matching: Why do people systematically overshoot when asked to reproduce a force applied to them? Explain using the forward model and sensory attenuation.

Exercise 3 — Clinical Application: Patients with schizophrenia CAN tickle themselves. What does this tell us about their forward model? How does this connect to auditory hallucinations (hearing voices that are actually internal speech)?

Write your response here

Part 5: Saccadic Eye Movements as Active Inference

Learning Goal: Analyze eye movements as proprioceptive predictions.

Exercise: Look at a natural scene photograph for 30 seconds. Then answer:

1. Where did your eyes go first? (Probably areas of high contrast, faces, text — areas of high expected free energy / information gain)
2. How long did fixations last? (Longer fixations = more information to extract = more prediction errors to resolve)
3. Did you revisit any locations? (Revisiting = renewed uncertainty about that region)
4. Where did you NOT look? (Low expected information gain — predictable regions like blank walls)

Now connect your observations to Active Inference: saccades are epistemic actions that direct the fovea toward regions of maximum expected free energy reduction.

Write your response here

Part 6: Free Will and the Readiness Potential

Learning Goal: Critically analyze the implications of the RP for Active Inference and volition.

Exercise: Read the following simplified description of Libet's experiment (1983):

"Participants were asked to flex their wrist whenever they wanted. EEG recorded a slow negative potential (the readiness potential, RP) beginning ~550ms before the movement. But participants reported conscious awareness of their 'intention to move' only ~200ms before the movement — 350ms AFTER the RP began."

1. What does the classical interpretation conclude about free will?
2. How does Active Inference reinterpret the RP? (Hint: the RP is gradual precision accumulation, not a "decision")
3. Does conscious awareness of intention need to precede the RP for free will to exist? Why or why not?
4. Write a 200-word argument for why the Libet experiment does NOT refute free will under Active Inference.

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Framework comparison	Predictions vs. commands
2	Cerebellar analysis	Forward model and error-driven learning
3	Circuit mapping	Basal ganglia policy selection
4	Experimental reasoning	Sensory attenuation and self-model
5	Behavioral analysis	Saccades as epistemic actions
6	Philosophical analysis	Readiness potential and volition